A DATA-DRIVEN MARKET STUDY · JULY 2026

Mapping the Subsea Tier-1.

A clustering study of the top 15 players.
Six strategic groups. Six ways of buying.

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Python · pandas · scikit-learn · scipy · numpy · matplotlib
Full article, code & data → brunovanham.github.io/subsea-clusters

EXECUTIVE SUMMARY

Subsea is not coming back. It is already back.

In the last ninety days alone, the industry awarded roughly USD 4.4 billion in subsea contracts: a supermajor SURF award in Brazil, a billion-dollar installation package in Angola, a cluster of tie-backs in Norway. Behind the awards sits a funded pipeline: Petrobras alone plans 180 new wells and 12 FPSOs in the pre-salt by 2030, and tie-back economics now put deepwater breakevens near USD 35 per barrel.

USD 4.4B	47	58%	USD 36B
Subsea awards Apr-Jul 2026	Offshore FIDs in 2025 up from 38	Of global EPCI value held by the top 5	Projected market size by 2034

The subsea market is not one single thing. There are vessel owners, equipment manufacturers, cable makers and diversified giants, and they do not bid for the same contracts. This study maps who actually competes with whom, within the tier-1 layer of the industry.

METHODOLOGY

The method, tool by tool

`pandas`

01 · Build the dataset

One row per company, nine features each: estimated subsea revenue, share of the business that is truly subsea, scope breadth (SPS, SURF installation, flexibles, cables, services), fleet size and Brazil presence. Sources: 2025 annual reports and market research. Every estimate is declared and editable.

`scikit-learn · StandardScaler`

02 · Standardize the features

Fleet ranges from 0 to 50 vessels; scope flags are 0 or 1. Without standardization the big numbers crush the small ones and the clustering is meaningless.

`scikit-learn · K-Means + silhouette`

03 · Let the algorithm find the groups

I did not tell it which groups exist. I tested k from 2 to 7 and the silhouette score picked k = 6.

`scipy · Ward linkage`

04 · Cross-check with a second method

Hierarchical clustering drew the same structure. Two independent methods, same map.

`numpy · distance decomposition`

05 · Explain the outliers

For every company sitting alone, decompose the distance to its nearest neighbour and name the variable responsible.

RESULTS

Six strategic groups. Six different ways of buying.

The K-Means algorithm identified six coherent groups within the tier-1 layer. The map below plots the top 15 players on the first two principal components: the horizontal axis captures size and scope breadth; the vertical axis, an installation-vs-services orientation. Bubble size reflects estimated subsea revenue.

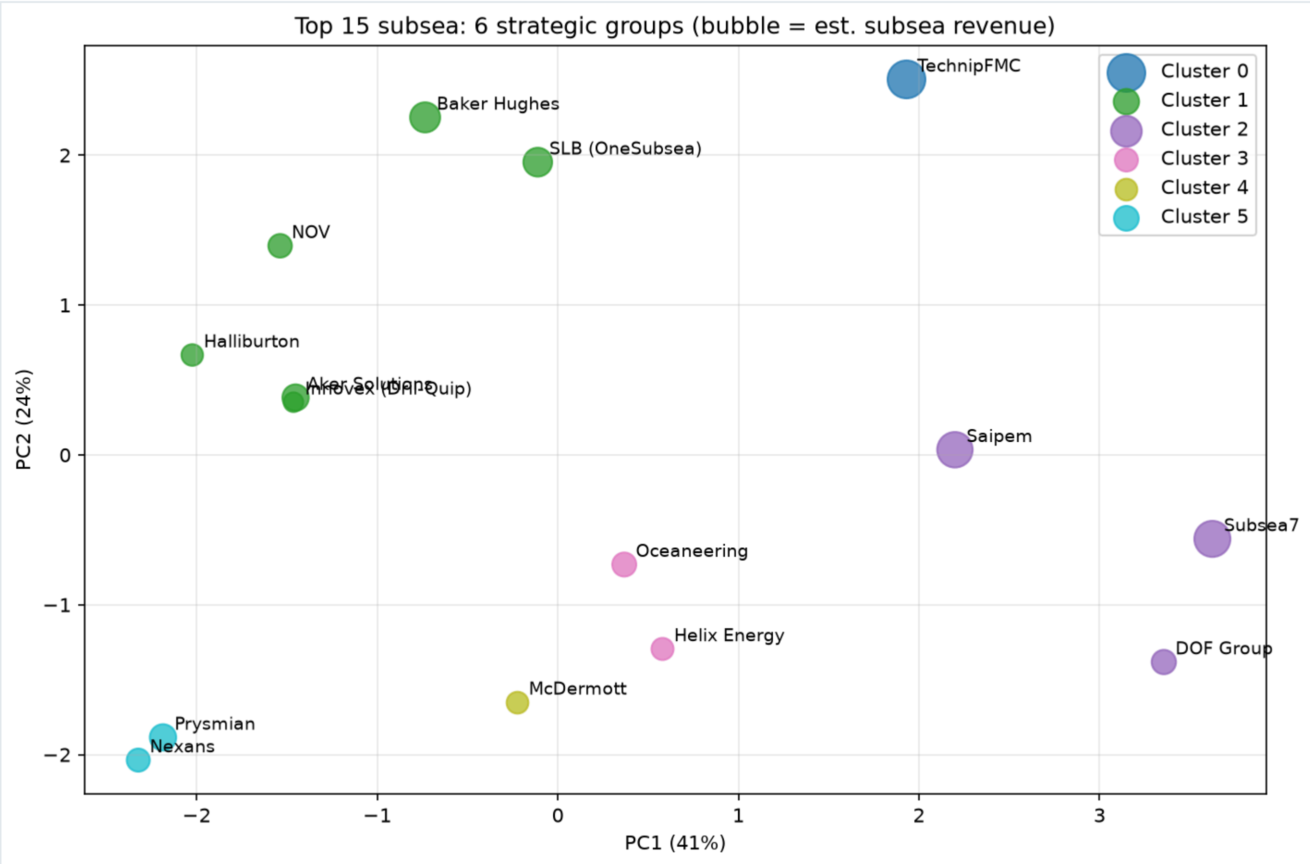

Figure 1. The competitive map of the subsea market, top 15 players. K-Means (k = 6, chosen by silhouette) projected via PCA.

KEY FINDINGS

What the algorithm actually said

The fleet owners: Subsea7, Saipem, DOF

Installation is king. The finding that most impressed me: the algorithm placed Subsea7 and Saipem as the two closest profiles among the top 15, the exact two companies merging into Saipem7 in 2026. The math and the M&A bankers reached the same conclusion, independently.

The lone integrated player: TechnipFMC

Isolated because it has too much: more than two standard deviations above the group in scope breadth, the only company that manufactures equipment, makes cables and installs. On this map, isolation means moat.

The opposite case: McDermott

Also alone, for the opposite reason. The distance decomposition shows an installer's profile without the scale of the fleet club. Same shape on the chart, opposite story.

The rest of the picture

Cable specialists (Prysmian, Nexans), services and intervention (Oceaneering, Helix) and the diversified giants (SLB, Baker Hughes, Halliburton, NOV, Aker, Innovex). Each group is a different buying center with different pain points and different procurement logic.

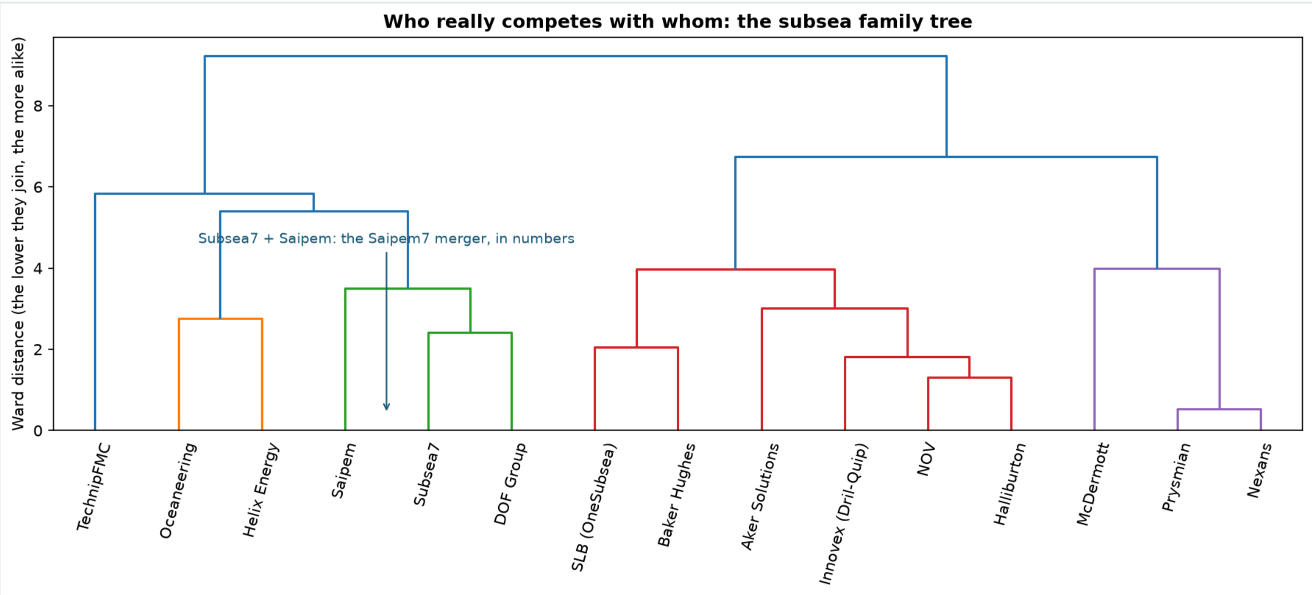

Figure 2. Hierarchical clustering (Ward linkage). Subsea7 and Saipem join at the very base of the tree: the Saipem7 deal, quantified.

VALIDATION

I tried to destroy my own study.

A clustering of fifteen companies deserves scrutiny, including mine. Before publishing, I stress-tested the model three ways. First, leave-one-feature-out: I removed each variable in turn and re-clustered. The groups held, with a mean Adjusted Rand Index of 0.73, well above the 0.5 threshold at which the structure would be feature-dependent. Second, Monte Carlo: I perturbed my own subsea-share and fleet estimates by ±20% across three hundred seeded simulations and measured how often each company stayed in its original group. Average stability: 90.8%. Third, I re-ran everything with Gower distance, the technically correct metric for a mixed matrix of continuous and binary variables. Same six groups.

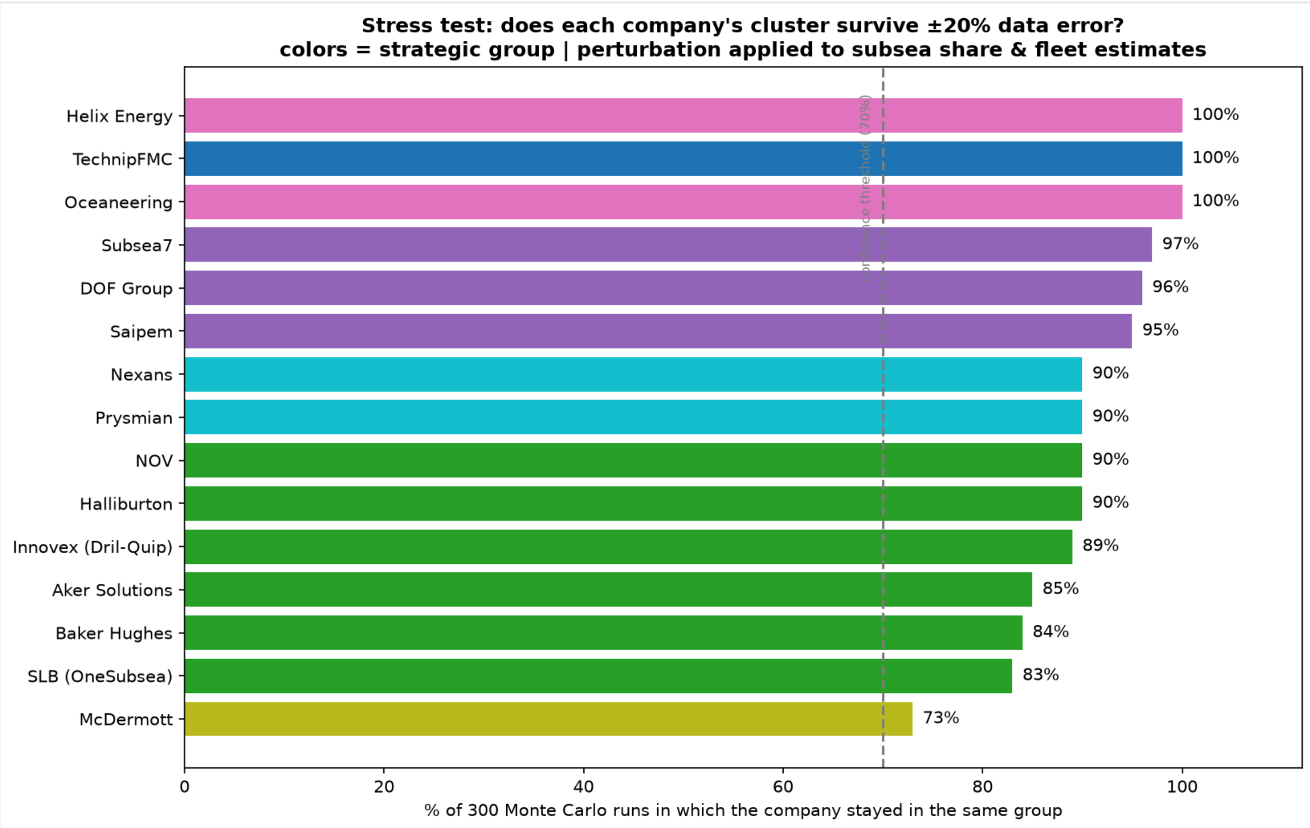

Figure 3. Monte Carlo stability per company (300 seeded runs, ±20% perturbation on estimated fields). Every company stays above the 70% threshold; the least stable, McDermott, still holds its group in 73% of runs.

ABOUT THIS STUDY

Honest limitations, deliberately declared

> *"The difference between a weak study and an honest one is not the absence of limitations. It is admitting them."*

This is a descriptive, exploratory analysis: it quantifies and organizes known industry similarities rather than discovering them blindly. It is also sensitive to the features chosen, which is exactly why the dataset is open. The scope is deliberately the tier-1 layer of the market. Next iterations will expand to specialists, regional players and the wider supply chain. If you have better numbers than my estimates, disagree with how a company was classified, or have insights the model should incorporate (real backlog composition, execution track record, technology depth, vessel utilization, standards and qualifications), the repository is open and contributions are welcome.

ABOUT THE AUTHOR

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SOURCES

Company press releases and 2025-26 annual reports (Subsea7, TechnipFMC, Saipem, Baker Hughes, Oceaneering). Mordor Intelligence, Subsea Systems Market 2026. Spherical Insights, Top 20 Subsea Companies 2026. Fortune Business Insights. DataM Intelligence.

Revenue figures are group-level and rounded; subsea share, fleet and scope flags are the author's declared estimates.

Full online article, interactive charts, code and data: **brunovanham.github.io/subsea-clusters**